

DATA FLOW DIAGRAMS (DFD)

AI Recruitment & Job Management System

1. Introduction

Data Flow Diagrams (DFDs) are graphical models used in software engineering to represent how data enters a system, how it is processed, where it is stored, and how outputs are delivered to external entities. DFDs focus on the logical flow of data rather than implementation details such as program code or user interface.

For the **AI Recruitment & Job Management System**, the DFD hierarchy models the flow of information between Candidates, Recruiters, and Administrators, along with the internal system processes, data stores, and external AI/NLP services. The terminology and functional decomposition used here are derived from the Software Requirements Specification (SRS) of the system.

2. What is a Data Flow Diagram?

A Data Flow Diagram (DFD) is a graphical representation of how data moves through an information system. It illustrates how data is processed, stored, and transferred between different components.

DFDs are organized into multiple levels, starting from a high-level overview (Level 0) and progressing to detailed representations (Level 1 and Level 2), enabling a clear understanding of system functionality.

3. Objectives of DFD

- Identify the sources and destinations of data in the system.
- Show how input data is transformed into meaningful output.
- Represent major processes and system boundaries.
- Identify data storage requirements.
- Improve communication among developers, analysts, and stakeholders.
- Support system design and requirement analysis.

4. Standard DFD Notations

Symbol	Meaning	System Example
Rectangle	External Entity	Candidate, Recruiter, Admin
Circle	Process	Resume Parsing, Job Matching, Application Tracking
Open-ended Rectangle	Data Store	Candidate Database, Job Database, Application Database
Arrow	Data Flow	Resume Data, Job Details, Match Score, Notifications

5. Important DFD Rules

- Every data flow must be clearly labeled.
- Processes must transform input data into output data.
- Data stores are used for persistent storage.
- External entities should not directly interact with data stores.
- Each process must have valid inputs and outputs.
- Level 2 diagrams must be a detailed decomposition of Level 1 processes.
- Lower-level DFDs must maintain consistency with higher-level diagrams.

6. DFD Levels Used for the AI Recruitment & Job Management System

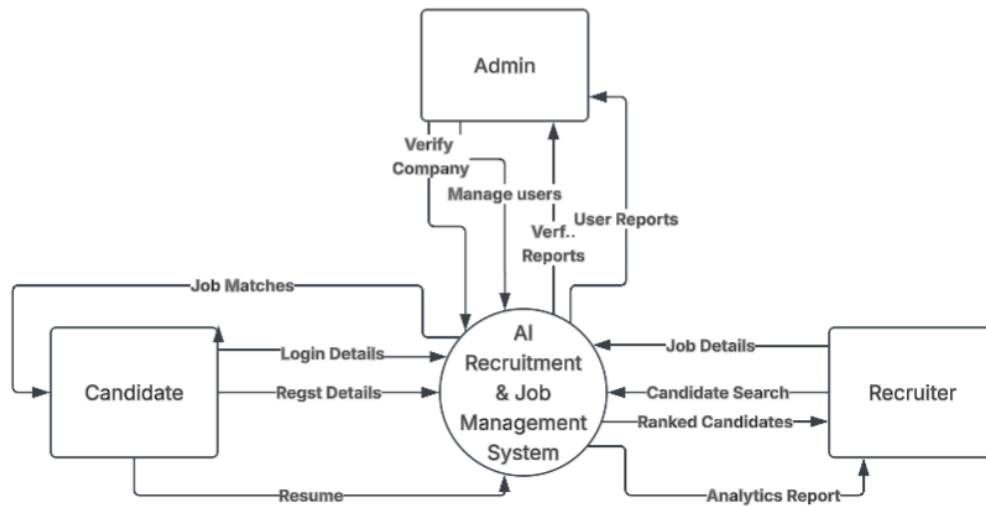
6.1 Level 0 – Context Diagram

The entire system is represented as a single process:
AI Recruitment & Job Management System.

This level shows interactions between:

- Candidate
- Recruiter
- Admin

Internal processes and data stores are not shown at this level.

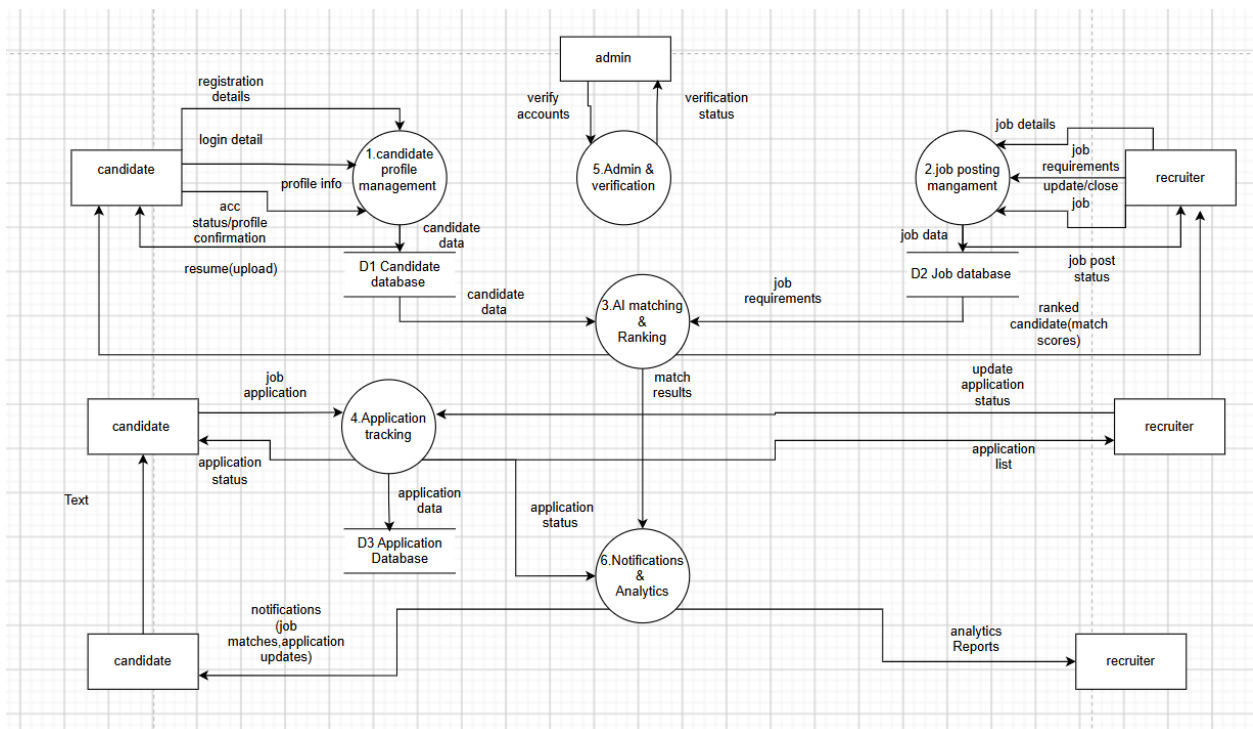
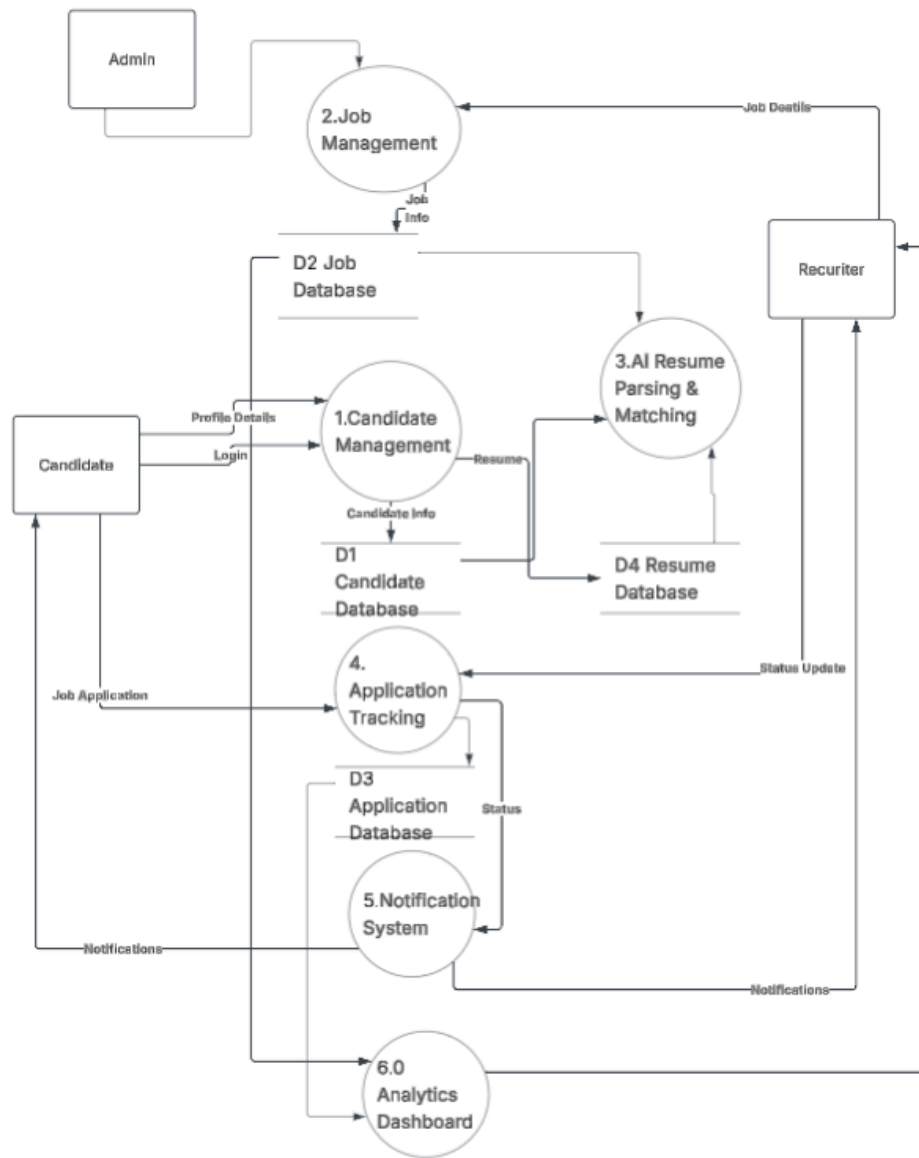


6.2 Level 1

Level 1 breaks the system into its major functional modules:

- Candidate Management
- Job Management
- AI Resume Parsing & Matching
- Application Tracking System (ATS)
- Notification System
- Reporting & Analytics

Each module represents a key functionality of the system and shows interactions with data stores and external entities.

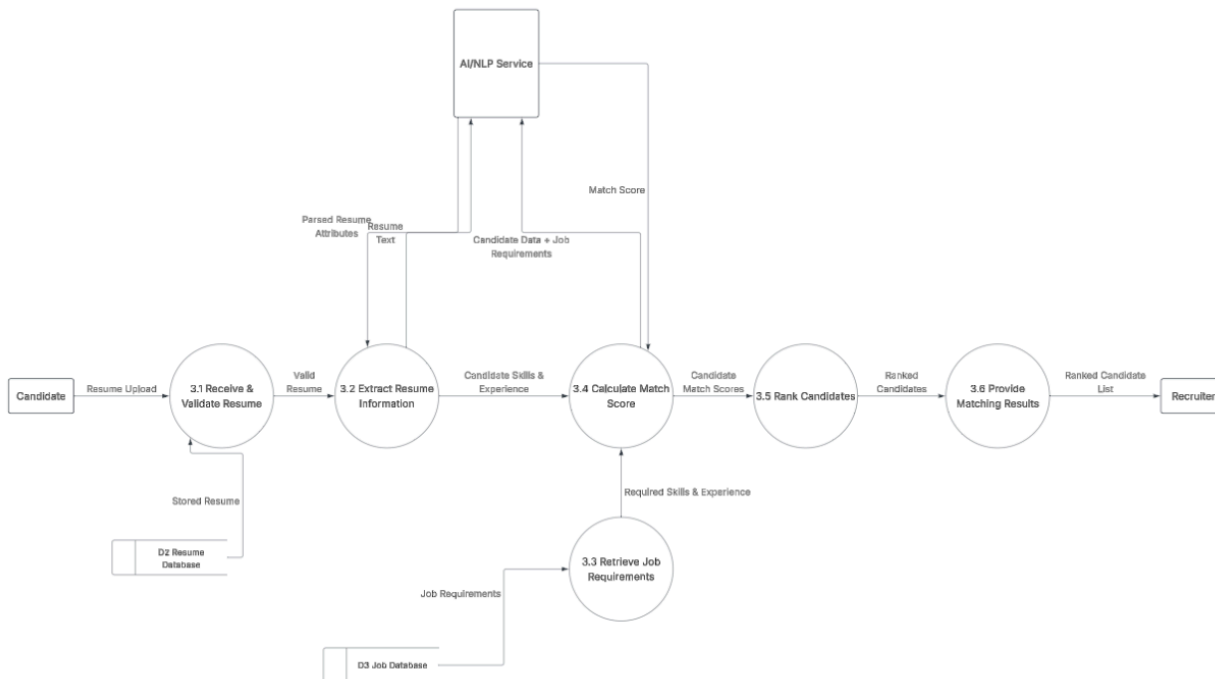


6.3 Level 2

Level 2 provides detailed decomposition of selected Level 1 processes.
The following Level 2 DFDs are included:

3.0 AI Resume Parsing & Matching

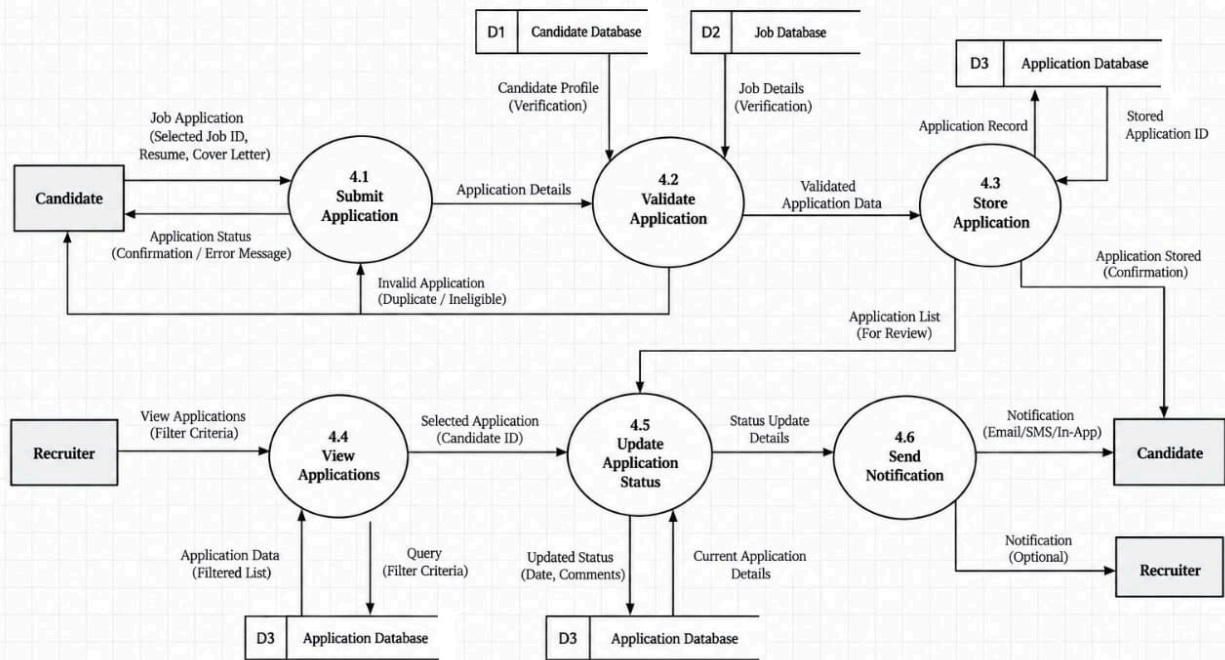
- Resume validation
- Information extraction
- Job requirement retrieval
- Match score calculation
- Candidate ranking



4.0 Application Tracking System (ATS)

- Application submission
- Validation
- Storage
- Status update
- Notification

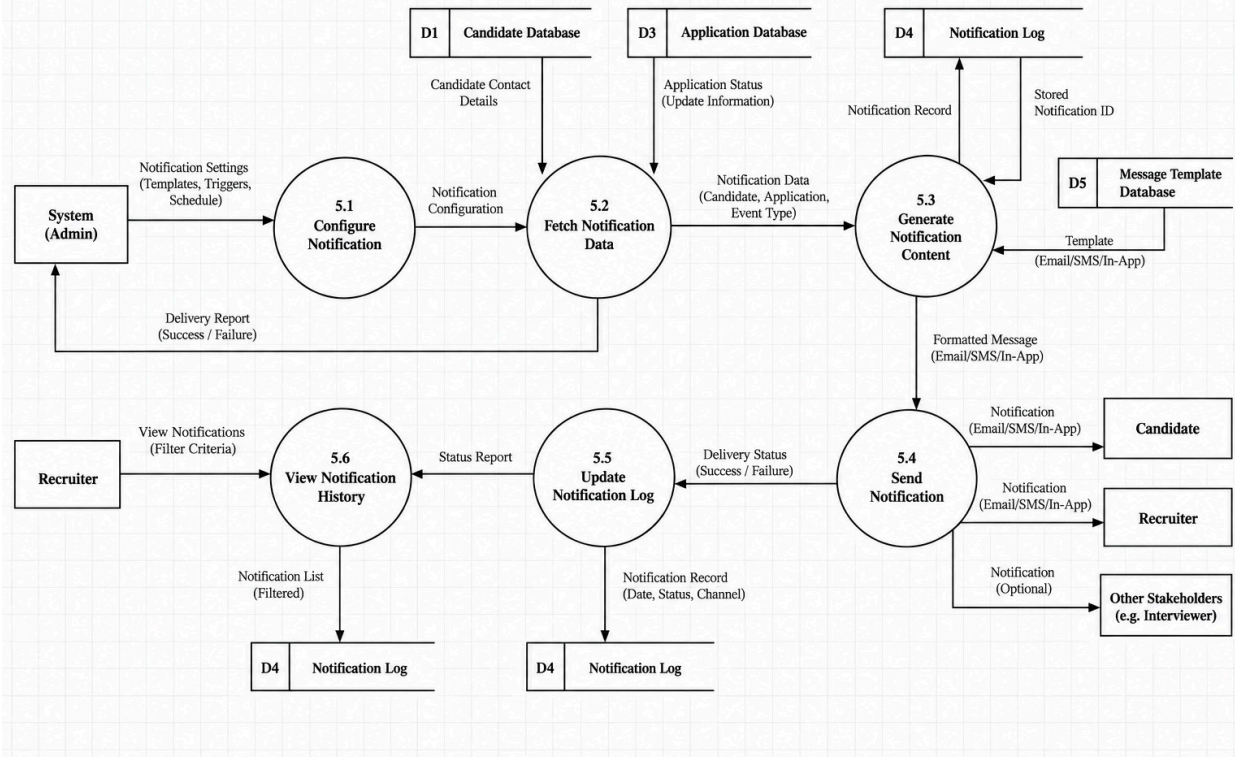
Level 2 DFD : 4.0 Application Tracking System (ATS) Process



5.0 Notification System

- Notification configuration
- Data fetching
- Content generation
- Sending notifications
- Logging and history

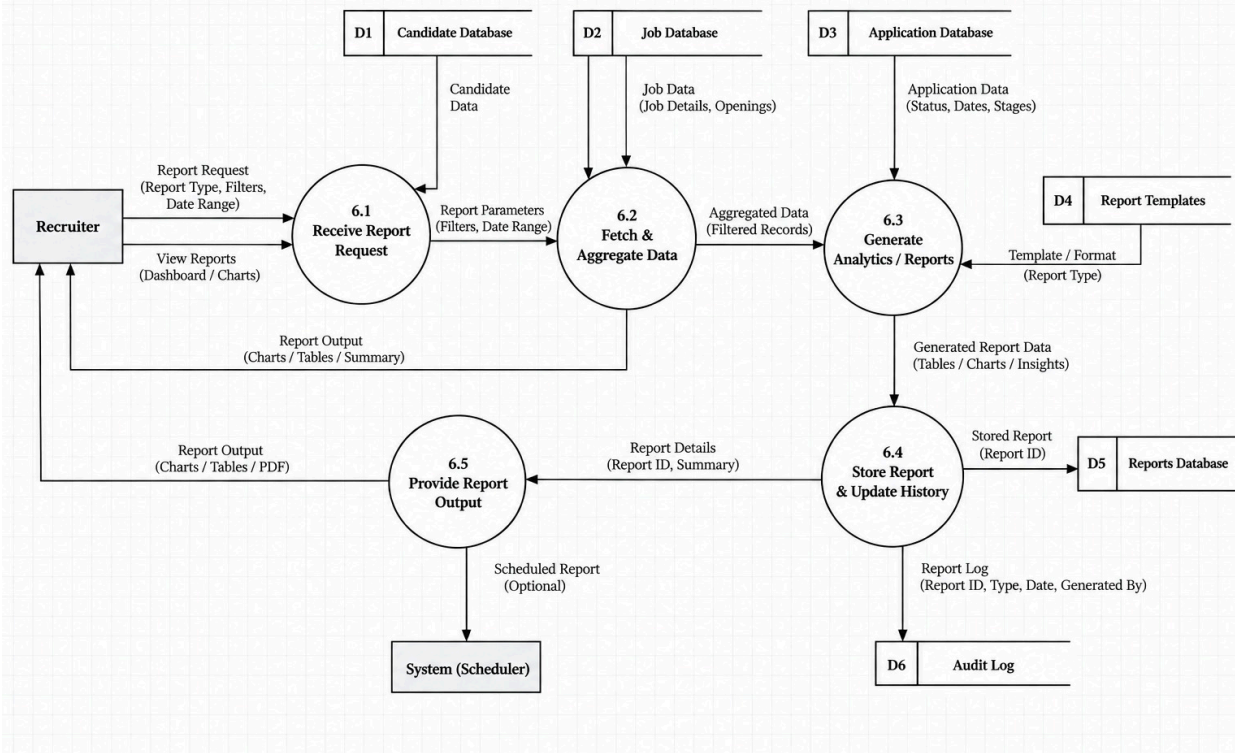
Level 2 DFD : 5.0 Notification System Process



6.0 Reporting & Analytics

- Report request processing
- Data aggregation
- Report generation
- Storage and audit logging
- Report delivery

Level 2 DFD : 6.0 Reporting & Analytics Process



Major Data Flows in AI Recruitment & Job Management System

The system consists of multiple key data flows that represent how information moves between external entities, processes, and data stores.

1. Candidate Registration & Login Flow

- Candidate → Registration Details → Candidate Management
- Candidate Management → Store Data → Candidate Database
- Candidate → Login Credentials → System
- System → Authentication Result → Candidate

2. Resume Upload & Processing Flow

- Candidate → Resume Upload → AI Resume Parsing Module
- Resume → Stored in → Resume Database
- Resume → Sent to → AI/NLP Service
- AI/NLP Service → Parsed Data → System

3. Job Posting Flow

- Recruiter → Job Details → Job Management
- Job Management → Store → Job Database
- Admin → Verification → Job Management

4. AI Matching Flow (CORE FLOW)

- Candidate Data + Resume → AI Matching Module
- Job Requirements → Retrieved from Job Database
- AI Module → Match Score → System
- System → Ranked Candidates → Recruiter

5. Job Application Flow (ATS)

- Candidate → Job Application → Application Tracking System
- System → Validate Application → Candidate/Job Database
- Valid Application → Stored in → Application Database
- System → Confirmation → Candidate

6. Application Status Update Flow

- Recruiter → Status Update → ATS
- ATS → Update → Application Database
- ATS → Status Notification → Candidate

7. Notification Flow

- Event Trigger (Status Change / Match Found) → Notification System
- Notification System → Fetch Data → Databases
- System → Generate Message → Template DB
- System → Send Notification → Candidate / Recruiter
- System → Log Notification → Notification Log

8. Analytics & Reporting Flow

- Recruiter → Report Request → Analytics Module
- Analytics → Fetch Data → Candidate, Job, Application DB
- System → Generate Report → Report Templates
- System → Store Report → Reports Database
- System → Display Output → Recruiter

9. Admin Control Flow

- Admin → Manage Users → System
- Admin → Verify Recruiters → Job Management
- System → Reports → Admin

10. Advantages of DFDs

- Provides a simple and clear visualization of data flow within the recruitment system.
- Helps understand how candidate data, job details, and applications move through the system.
- Improves communication between developers, stakeholders, and recruiters.
- Helps identify missing processes such as validation, matching, and notification.
- Supports requirement analysis and logical system design.
- Acts as a foundation for designing databases (Candidate, Job, Application, Notification).
- Helps testers understand system workflow, including application tracking and status updates.
- Supports hierarchical representation from high-level system overview to detailed modules like AI matching and analytics.

11. Limitations of DFDs

- DFDs do not describe internal algorithms such as AI matching logic or scoring methods.
- They do not represent user interface details like dashboards or forms.
- Complex systems like recruitment platforms can lead to cluttered diagrams.
- Timing aspects such as real-time notifications or processing delays are not shown.
- Control flow, concurrency, and error handling are not explicitly represented.
- DFDs should be complemented with other diagrams such as ER diagrams, Use Case diagrams, and Sequence diagrams.

Conclusion

The Data Flow Diagram (DFD) hierarchy provides a structured and logical representation of the **AI Recruitment & Job Management System**.

The **Level 0 context diagram** defines the system boundary and interactions with external entities such as Candidates, Recruiters, and Admins.

The **Level 1 diagram** decomposes the system into major functional modules including Candidate Management, Job Management, AI Resume Parsing & Matching, Application Tracking System (ATS), Notification System, and Reporting & Analytics.

The **Level 2 diagrams** further break down key modules such as AI Matching, ATS, Notification System, and Analytics into detailed subprocesses.

Together, these diagrams provide a comprehensive view of how the system collects candidate data, processes resumes using AI, manages job applications, sends notifications, and generates analytical reports. This structured representation helps in better system design, implementation, and maintenance.